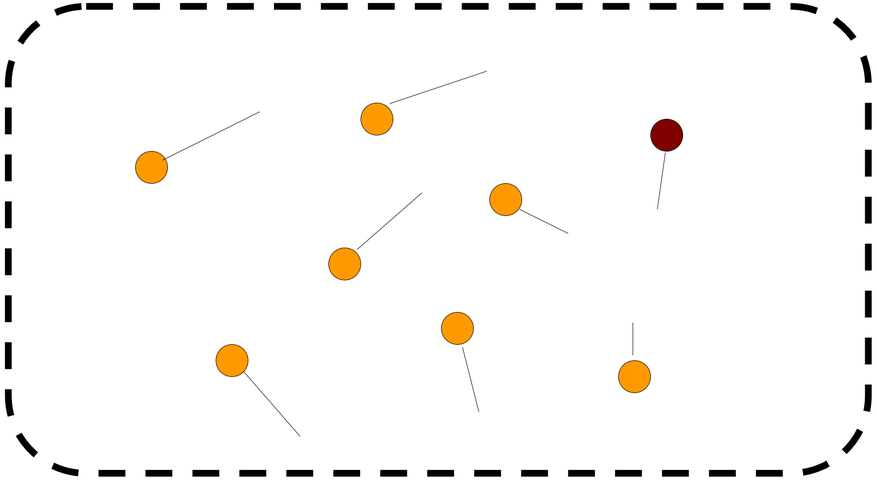
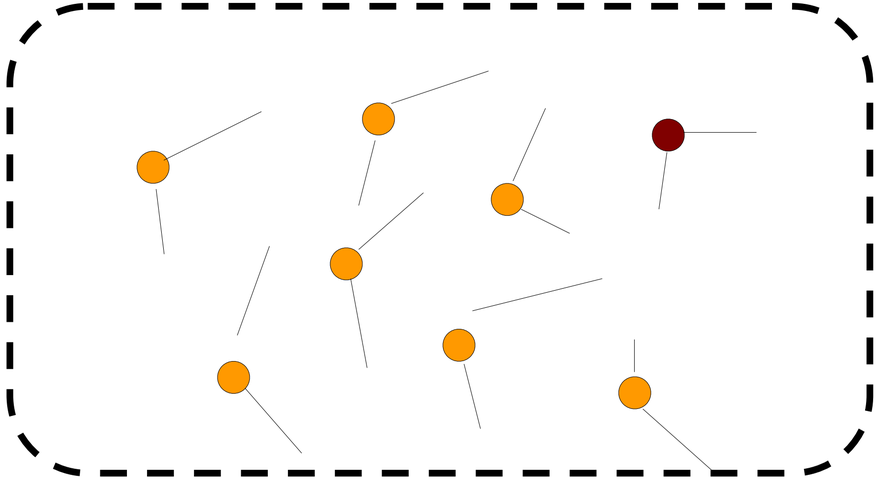
Homogeneous MF assumption



Class of individuals with degree k=1



Class of individuals with degree k=2

Degree based representation

MOBS LAB

Dynamic equations

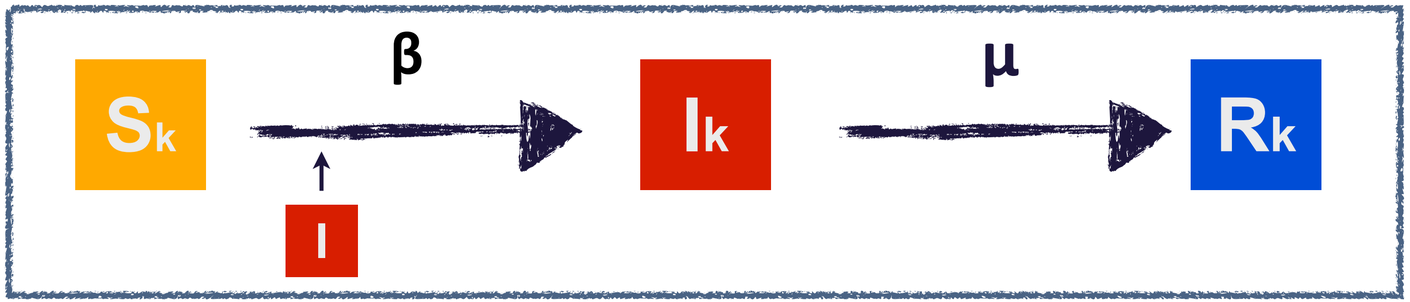
$$S_k(t+1) = S_k(t) - \boxed{\beta S_k(t)\frac{I(t)}{N}}$$

$$I_k(t+1) = I_k(t) + \boxed{\beta S_k(t)\frac{I(t)}{N}} - \mu I_k(t)$$

$$R_k(t+1) = R_k(t) + \mu I_k(t)$$

Homogenous interactions???

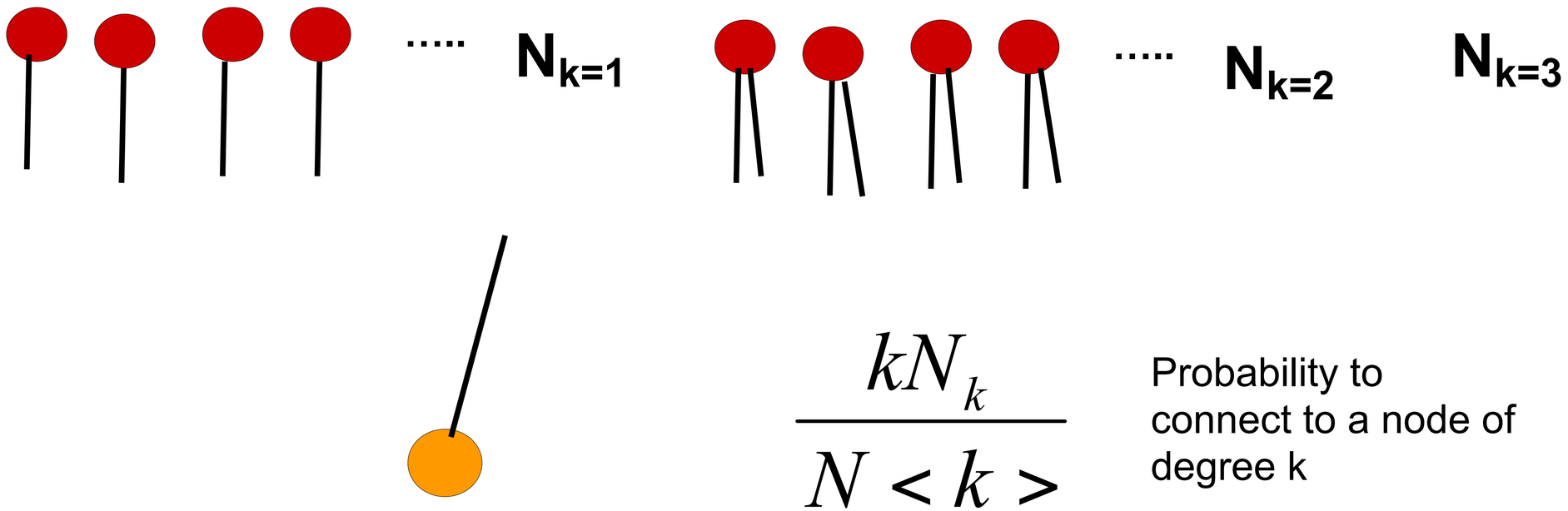
Average dynamic



Interactions with all possible class k

$$p_{S_k \to I_k} = \lambda \delta t k \sum_{k'} k' \frac{P(k) I_k(t)}{< k > N'_k}$$

What is the probability that this link points to an infected individuals?



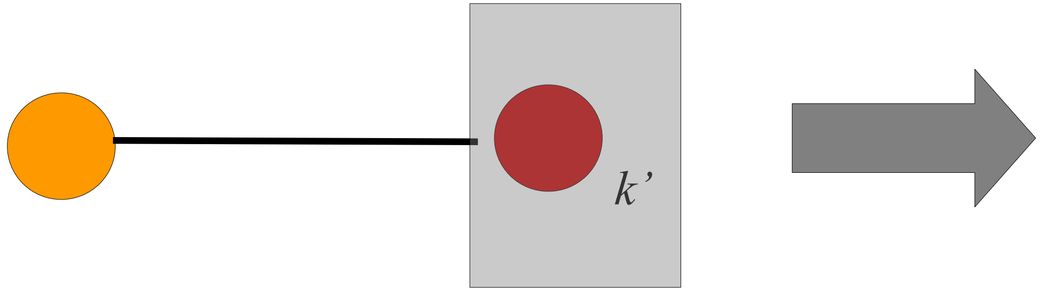
$$\frac{kN_k}{N < k >}$$

Probability to connect to a node of degree k

Average on all degree *k*

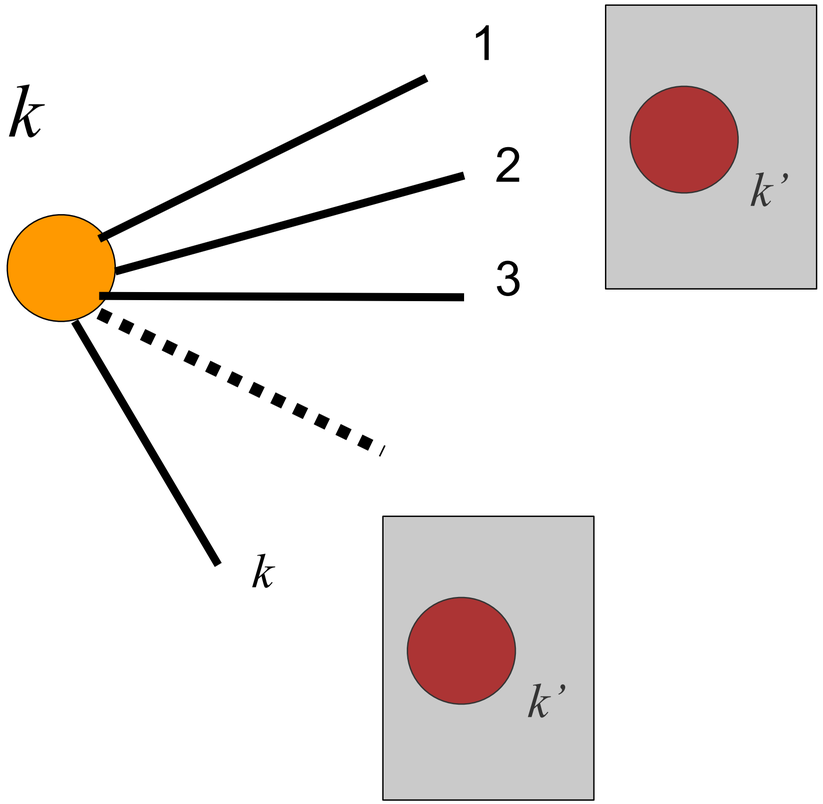
$$\frac{k' N_{k'}}{N < k >} = \frac{k' P(k')}{< k >}$$

All possible k’



$$\sum_{k'} \frac{k' P(k')}{< k >} \frac{I_{k'}}{N_{k'}}$$

Each susceptible of degree k has k possible connections



$$P_{s\to I} = 1 - (1 - \lambda \sum_{k'} \frac{k' P(k')}{< k >} \frac{I_{k'}}{N_{k'}})^k$$

OR

$$P_{s\to I} \approx \lambda k \sum_{k'} \frac{k' P(k')}{< k >} \frac{I_{k'}}{N_{k'}}$$

$$S_{k,t+1} = S_{k,t} - Binom(S_{k,t}, \boxed{p_{S_k \to I_k}})$$

$$I_{k,t+1} = I_{k,t} + Binom(S_{k,t}, \boxed{p_{S_k \to I_k}}) - Binom(I_{k,t}, \mu\Delta t)$$

$$R_{k,t+1} = R_{k,t} + Binom(I_{k,t}, \mu\Delta t)$$

$$S_{k,t+1} + I_{k,t+1} + R_{k,t+1} = S_{k,t} + I_{k,t} + R_{k,t}$$

constant population

$$I_k(t+1) = I_k(t) - \mu I_k(t) + \lambda S_k k \sum_{k'} \frac{k' P(k')}{\langle k \rangle} \cdot \frac{I_{k'}(t)}{N_{k'}}$$

$$\Theta(t) \equiv \sum_{k'} \frac{k' P(k')}{\langle k \rangle} \cdot \frac{I_{k'}(t)}{N_{k'}}$$

$$I_k(t+1) = I_k(t) - \mu I_k(t) + \lambda S_k k \, \Theta(t)$$

$$\sum_k \frac{k P(k)}{\langle k \rangle N_k} I_k(t+1) = \sum_k \frac{k P(k)}{\langle k \rangle N_k} I_k(t) - \sum_k \frac{k P(k)}{\langle k \rangle N_k} \mu I_k(t) + \sum_k \frac{k P(k)}{\langle k \rangle N_k} \lambda S_k k \, \Theta(t)$$

$$\sum_k \frac{k P(k)}{\langle k \rangle N_k} I_k(t+1) = \Theta(t+1)$$

$$\sum_k \frac{k P(k)}{\langle k \rangle N_k} I_k(t) = \Theta(t)$$

$$\sum_k \frac{k P(k)}{\langle k \rangle N_k} \mu I_k(t) = \mu \Theta(t)$$

$$\sum_k \frac{k P(k)}{\langle k \rangle N_k} \lambda S_k k \Theta(t) = \lambda \Theta(t) \sum_k \frac{k^2 P(k)}{\langle k \rangle N_k} \cdot S_k$$

Early epidemic $S_k \simeq N_k \Rightarrow \lambda \Theta(t) \frac{\langle k^2 \rangle}{\langle k \rangle}$

(linear approx)

Closed equation for $\Theta(t)$

$$\Theta(t+1) = \Theta(t) - \mu\,\Theta(t) + \lambda \frac{\langle k^2 \rangle}{\langle k \rangle}\,\Theta(t)$$

$$\Theta(t+1) = \left[\frac{\lambda \langle k^2 \rangle}{\langle k \rangle} - \mu + 1 \right] \cdot \Theta(t)$$

Condition for a growing epidemic:

$$\frac{\lambda \langle k^2 \rangle}{\langle k \rangle} - \mu + 1 \geq 1$$

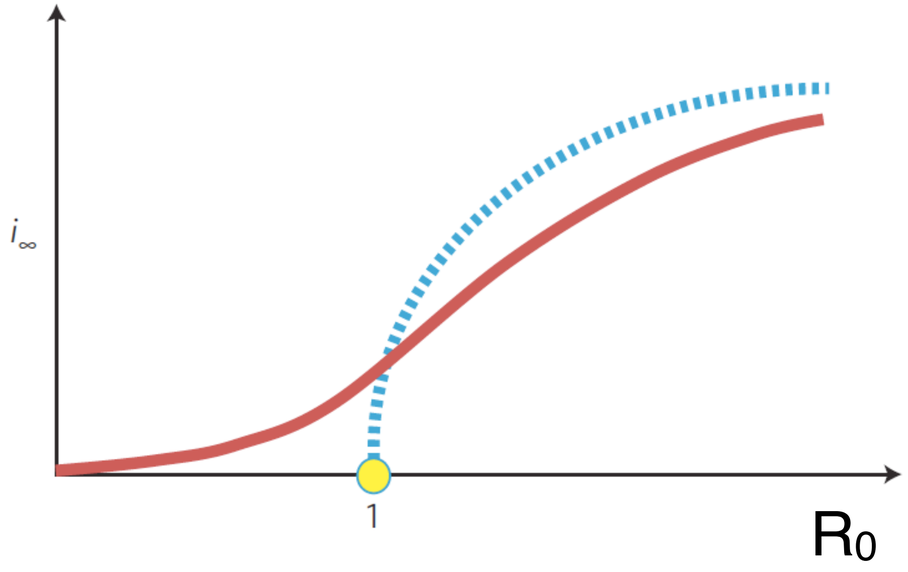
$$\frac{\lambda \langle k^2 \rangle}{\mu \langle k \rangle} \geq 1 \qquad \text{if } \beta = \langle k \rangle \lambda \implies \frac{\beta}{\mu} \frac{\langle k^2 \rangle}{\langle k \rangle^2} \geq 1$$

NETWORK TOPOLOGY AFFECTS EPIDEMIC THRESHOLD AND DYNAMICAL BEHAVIOR OF EPIDEMICS

•Network heterogeneity tend to suppress the spreading threshold

- **Random-annealed network patterns have a vanishing epidemic threshold**

$$R_0 \rightarrow R_0 \langle k^2 \rangle / \langle k \rangle^2$$

•Quenched networks have an epidemic threshold vanishing as the inverse of the largest eigenvalue of the connectivity matrix.

•Structure may change the results, however, assortativity/disassortativity do not impact the qualitative behavior concerning threshold.